IQ Hack 2026 - Tunneling to the Top Report

Team Details

Team Name: Tunneling to the Top

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Submission Category: QML

Bulleted Summary

1. We brainstormed possible QML projects. We decided on applying the provided QML algorithm from the hackathon notebooks to a new dataset, and attempting to apply one other QML algorithm that we would find through research.

2. We looked into possible datasets, and decided on one with some interesting data on objects in Earth's orbit.
    - The CSV contains 40 features, 11 of which were orbital measurements. We limited ourselves to 4 features for easier application of QML algorithms. Our target variable categorizes the material into either debris (space junk) or payload (objects designed to perform a specific function).
    - We also capped our dataset at 300 total rows, split into training, validation, and testing sets, for timeliness when fitting models.

3. We applied the provided QSVC model to our data, editing the number of qubits to match our features.

4. To see if a difference in success between the quantum model and a classical classifier algorithm was visible, we also fit a classical Support Vector Machine model to our data.

5. We then did some research on Variational Quantum Classifiers before coding and fitting one to our data.

6. Comparing the validation results from each of the models, we found that the QSVC performed best and we ran it on our testing data to get a final accuracy score of 86.67%

7. As we worked on the supervised QML models, we became interested in the possible application of unsupervised machine learning to our dataset. We wondered if, even without the labels, the model would end up with similar groupings.
    - To test this, we implemented a classical k-means model to explore the possibility.
    - Some interesting results were found, and we discovered a small grouping within the data that held all the high Earth orbit data. When we allowed for 3 groups, the groupings other than the high Earth orbit group agreed much more with our known labels.
    - Then we applied quantum kernel-based spectral clustering to the same data, to see if it performed any better. With only two clusters, the quantum algorithm did perform better than the two-cluster classical system.
8. To check how robust our results were, we built a second, filtered version of the data and tested the quantum kernel further.
    - We filtered the data to low Earth orbit only, removed large constellations like Starlink, and compared the QSVC with tuned classical models over 10 random samples. The scaling of the data (the bandwidth) decided whether the quantum kernel worked.
    - We tested the quantum kernel under shot noise, simulated IBM device noise, and on real IBM hardware (ibm_kingston). Accuracy stayed the same in all cases.
    - We applied the quantum kernel to a new problem: finding which of four breakup events a debris fragment came from.

Motivation

The team chose to run QML algorithms on orbital debris data from Space-Track because the possibility of collision with space debris is increasingly a problem for the space industry. It is a costly and data heavy area within the field, and the team was genuinely curious about whether quantum advantage could be found in this arena.

The World Economic Forum estimates that debris could cause between $25.8 and $42.3 billion in cumulative losses between 2025 and 2035 [1], and NASA has published cost benefit analyses comparing more than 10 strategies for reducing collision risk, including deorbiting spacecrafts and actively removing large debris [2].

Countries, companies, and special interest groups are also paying for avoidance. Starlink alone performed 25,299 collision avoidance maneuvers in just six months [3].

Finally the Space-Track data catalog is structured and clean, so it let us test the hackathon notebook provided algorithm for quantum advantage without data-prep noise muddying the results:

Code Workflow

Phase 1: Environment and Data Pipeline
The notebook installs all necessary packages. Data is uploaded into the notebook and converted into a DataFrame of 300 orbital objects, 150 debris, and 150 payloads. The code verifies that the four model inputs (INCLINATION, ECCENTRICITY, MEAN_MOTION, BSTAR) and the binary label column exist. It will stop if any is missing with a clear error message. A 60/20/20 split is used, separating out 180 training objects, 60 validation objects, and 60 test objects. A MinMaxScaler scales each feature to the range [0, π] as appropriate for the rotation-angle representation in the quantum circuits. No information from the validation or test sets is used for preprocessing and so scaling is applied to the training set exclusively and without modification to the held out sets.

Phase 2: Quantum Kernel Classification (QSVC)
The first supervised model is a Quantum Support Vector Classifier model from the notebooks used in the hackathon. Each object is represented by a quantum state in a four-qubit ZZ feature map with two repetitions. A FidelityStateVectorKernel computes the overlap between each pair of states; the overlap is used as the kernel for the classifier. Simulator fitting and scoring only takes about 6 seconds. The accuracy of QSVC is 97.78% in the training set and 88.33% in the validation set.

Phase 3: Classical Baseline
A classical kernel is produced under the same conditions, as the training is performed on the same scaled features as well as the same splits with a classical support vector machine using a RBF kernel. It achieves a score of 83.33% on the validation set, five percentage points behind the QSVC.

Phase 4: Variational Quantum Classifier
The second quantum model is a Variational Quantum Classifier, constructed based on the Qiskit tutorial by Zaw Myint Tun [4]. The data is encoded on a ZZ feature map of two repetitions and linear entanglement. The trainable rotation and entangling layers are provided by a 4-qubit, 2-repetition RealAmplitudes ansatz. The model is built using a StatevectorSampler, cross entropy loss and the SPSA optimizer, both learning rate and perturbation are set to 0.1 and the number of iterations is 150. A callback is used to record the loss and print progress periodically. The model fitting takes about 8 minutes on a laptop and gives 73.33% accuracy on the

validation set. Fit and scoring calls are commented out in the notebook, so that the program runs fast, but they can be uncommented when needed.

Phase 5: Held-Out Test Evaluation

The QSVC model is the model with the best validation accuracy among the three supervised models (88.33%, compared to 83.33% for the classical SVM and 73.33% for the VQC) and it is therefore taken to final testing. The test set is only scored after model selection, and does not influence model selection. The QSVC is able to achieve an accuracy of 86.67%.

Phase 6: Unsupervised Exploration using k-means

The last experiments do not use the labels, but instead determine if the data falls into categories on its own. The 300 objects are all scaled to [0, 1] and are clustered using classical k-means (2 or 3 clusters). A crosstab by OBJECT_TYPE indicates that the debris and payload groups are not split completely into the different clusters. The clustering also reveals that there is an outlier group which is characterized by its high Earth orbit that is being singled out by the k-means algorithm. A graph displaying inclination vs. mean motion with each cluster of objects color-coded and debris shown as circles, payloads as triangles is created.

Phase 7: Quantum-Kernel Spectral Clustering and Classical Comparison

Quantum-kernel spectral clustering is done and compared to the classical results.
The same question is then asked of the quantum kernel. All 300 objects are scaled to [0, π] and the FidelityStateVectorKernel results in a 300 × 300 similarity matrix. This precomputed matrix is used for spectral clustering with two clusters. Since the numbering of clusters is arbitrary, an agreement score is the maximum of the forward and backward match rates with the correct clusters. The quantum kernel has a score of 0.717. The default gamma setting of a classical RBF kernel on the same scaled features and clustered as above achieves a score of 0.517. In the ideal case of balanced classes, this metric can't be less than 0.5, which is the minimum. To rule out an unfavorable kernel setting, the classical kernel is re-evaluated at seven gamma values (0.05, 0.1, 0.25, 0.5, 1, 2, 5). The agreement is between 0.517 and 0.523, and does not rely on the classical gamma choice for this experiment, showing the advantage of the quantum kernel.

Phase 8: Filtered Data and Fair Comparison

We loaded the full Space-Track catalog [5] (32,383 objects) and kept only objects in low Earth orbit, removing large constellations like Starlink. Each run still uses 300 objects (150 debris, 150 payloads) with the same *60/20/20 split*. Both the QSVC and the classical models (SVM, random forest, logistic regression) were tuned, and we repeated everything over 10 random samples, reporting the average test accuracy.

Phase 9: Noise Simulation

On a smaller set of 40 training and 20 test objects, we computed the quantum kernel three ways: ideal, with shot noise *(2,000 shots)*, and with the noise of a real IBM processor (FakeTorino).

Phase 10: IBM Quantum Hardware

We ran the same kernel on ibm_kingston [6], a real 156-qubit IBM processor, with 500 shots per circuit and error suppression turned on, and compared it with the ideal kernel.

Phase 11: Debris Breakup Attribution

As a new application, we used the quantum kernel to identify which of four breakup events a debris fragment came from (Fengyun-1C, Cosmos 2251 and two Long March 6A rocket stages). We used *60 fragments per event* and compared the QSVC with the same classical models over 6 random samples.

Results

Supervised Machine Learning: Comparing Quantum and Classical

The Quantum Support Vector Classifier (QSVC), as provided in the Interstate Quantum Hackathon notebooks, running with 4 qubits to match our data's four features returned a validation accuracy of 88.33%. The Classical Support Vector Machine Classifier (Classical SVM) returned a validation accuracy of 83.33%. The Variational Quantum Classifier, using a Simultaneous Perturbation Stochastic Approximation optimizer, following the design outlined in Zaw Myint Tun's Quantum Machine Learning Basics article [4], resulted in an accuracy of 73.33%, and required about 8 minutes of computing time. With these results, it's clear that the QSVC model performed best. On our set aside test data, the QSVC model resulted in 86.67% accuracy.

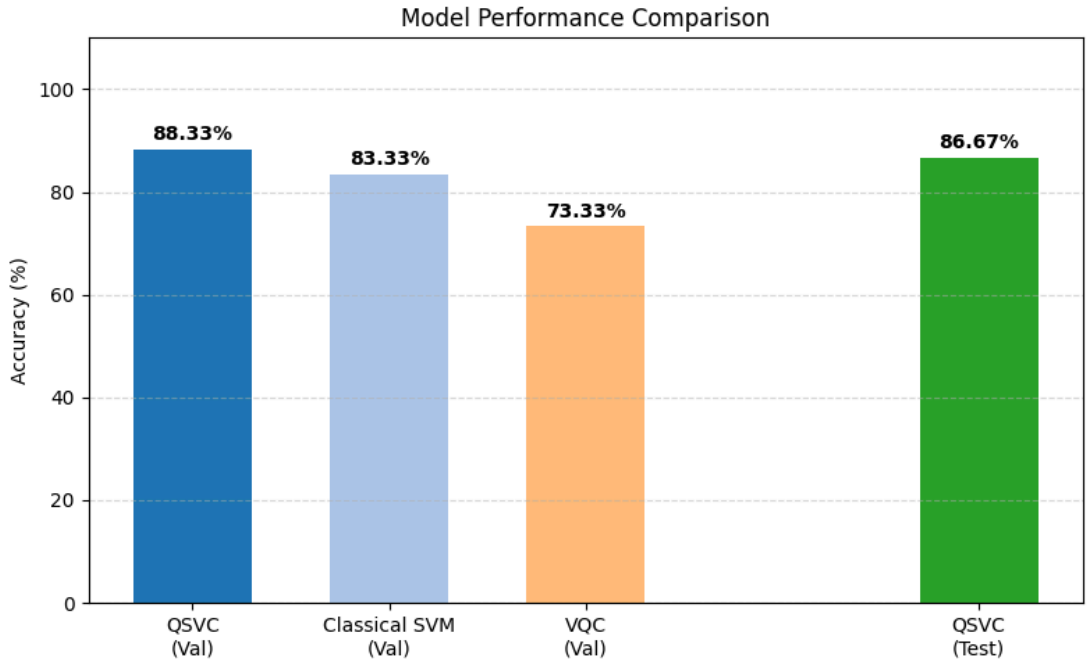


Fig 1. Comparing Supervised Model Performance

These results show some benefit to using quantum computing for this ML application, however small. Though our best performing model is a quantum ML model, our worst performing model is as well. This is not particularly surprising, as our data set isn't complex enough to be uninterpretable by classical machine learning methods and has no inherent quantum tendencies that would be best depicted by a matching QML model. All things considered, we successfully applied the QML model to a new dataset, and found and applied a second QML model to the same data, so we met our initial goals.

Unsupervised Exploration: Can We Recover Payload vs Debris Without Labels?

Our goal was to hide the payload and debris labels from the 300 object orbital dataset (150 debris and 150 payloads chosen at random) and see whether an unsupervised algorithm would rediscover the same groupings. Agreement is scored by counting the majority object type in each group as that group's answer.

We first ran classical k-means with two groups on four features: inclination, eccentricity, mean motion, and BSTAR. The result was not what we expected. k-means split off 15 high altitude objects (mean motion of about 1 lap per day), leaving agreement at 52.3%.

col_0	0	1
OBJECT_TYPE		
DEBRIS	4	146
PAYLOAD	11	139

Fig 2. Crosstab of 2 cluster k-means results

When we allowed a third group, agreement rose to 73.7% (221 of 300). With three groups, k-means sorted mainly by orbit tilt and altitude. Debris leaned toward steeply tilted orbits near 98°, while payloads were more spread out, and the third group was again the high altitude clump.

col_0	0	1	2
OBJECT_TYPE			
DEBRIS	118	28	4
PAYLOAD	47	92	11

Fig 3. Crosstab of 3 cluster k-means results

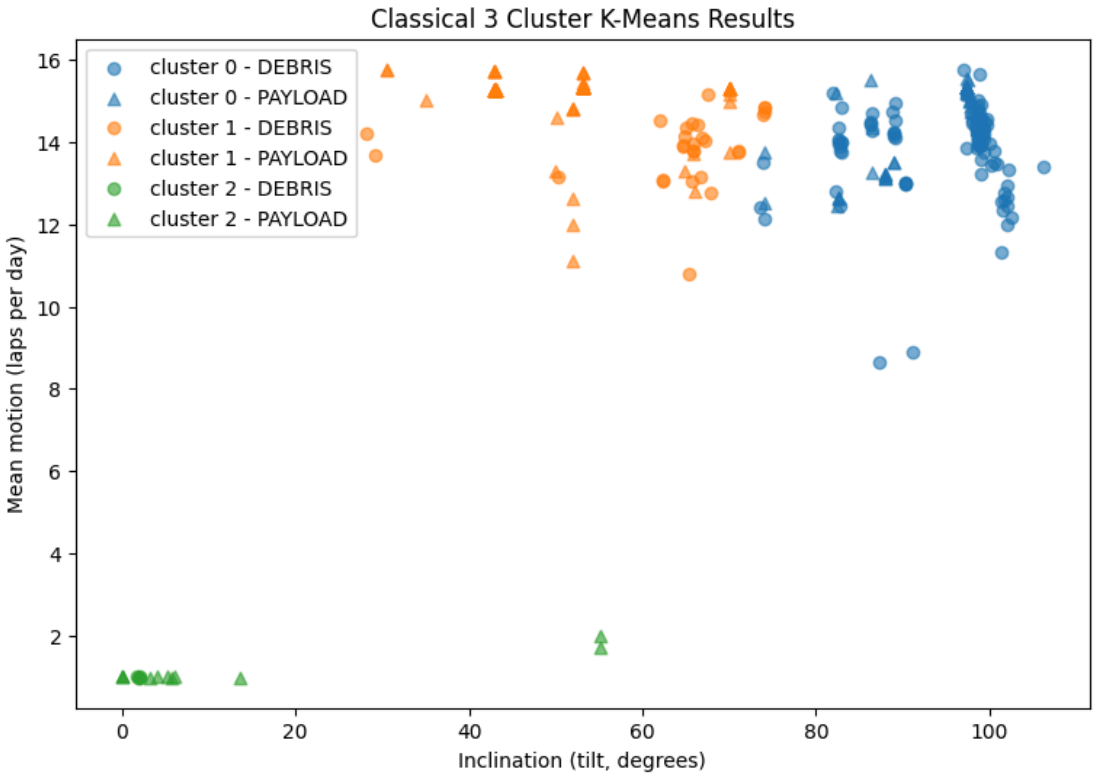


Fig 4. Classical 3 Cluster k-means Results, colored by cluster, shape by object type

We then built a quantum similarity table using the same feature map as our QSVC and clustered it with spectral clustering. Only the similarity measure is quantum, and the sorting step is classical. With two groups, the quantum kernel reached 71.7% agreement (215 of 300). Using the same clustering method with a classical RBF kernel gave only 51.7%, and it stayed between 51.7% and 52.3% across seven different gamma settings. So the quantum kernel recovered the labels about 20 points better than the classical kernel at two groups.

Comparing all of the unsupervised results, agreement with the true labels ranged from 51.7% to 73.7%. Classical k-means with two groups (52.3%) and spectral clustering with a classical kernel (51.7% to 52.3%) were close to chance, since random guessing on balanced classes gives about 50%. Classical k-means with three groups scored highest at 73.7%, just ahead of the quantum kernel with two groups at 71.7%, a gap of six objects out of 300. Since the three group method was given an extra cluster, the fairest comparison is among two group methods, where the quantum kernel beat both classical approaches by about 20 points. We therefore consider quantum kernel clustering the best unsupervised method when the number of groups is equal, with classical three group k-means a close second overall. These results come from a single run on 300 objects and one classical kernel type, so they are suggestive rather than conclusive.

Robustness Check: Filtered Data and Fair Comparison

To see whether the QSVC's advantage would hold up, we repeated the supervised comparison on the filtered data (low Earth orbit only, no large constellations), with every model tuned and the results averaged over 10 random samples. The tuned QSVC reached 89.8% (± 4.9%) test accuracy, compared with 91.0% for the tuned classical SVM, 89.0% for logistic regression and 88.8% for random forest. All four models fall within each other's error bars, so the quantum kernel is competitive, but it does not beat well-tuned classical models on this task.

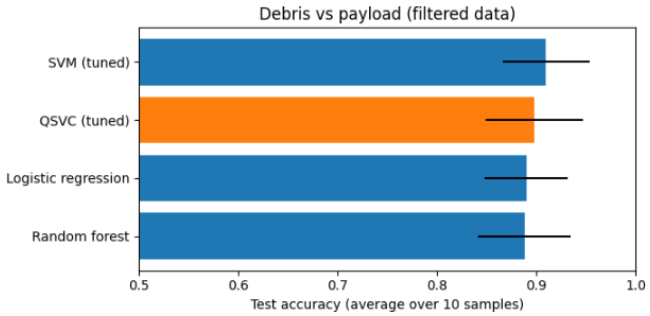


Fig 5. Test accuracy on filtered data

Why the Bandwidth Matters

The bandwidth controls how spread out the data is when it goes into the quantum circuit. In Part 1 we used λ = 1, and at that setting the accuracy drops to 64.8%, because the quantum kernel sees almost every pair of objects as completely different. With λ = 0.1 the accuracy rises to 87%. This was the most important setting for the quantum model.

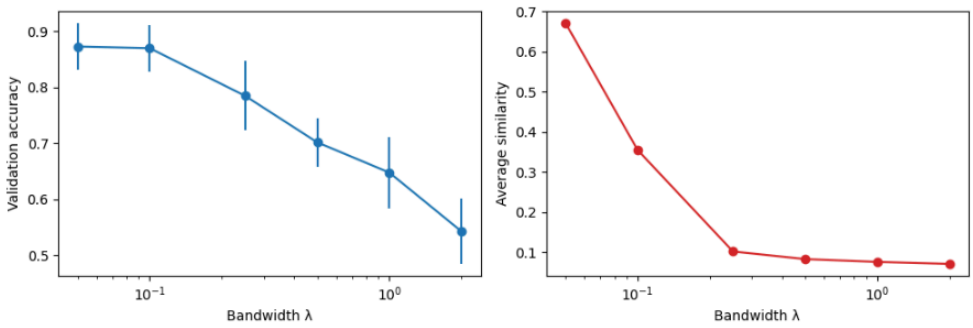


Fig 6. Accuracy and average similarity for different bandwidths

Testing with Noise

Real quantum computers are noisy, so we tested the same quantum kernel three ways on 40 training and 20 test objects: ideal (no noise), with shot noise (2,000 shots), and with the noise of a real IBM processor (FakeTorino). All three gave the same test accuracy of 80%. The noise changed the kernel values slightly (by 0.08 on average), but it did not change the predictions.

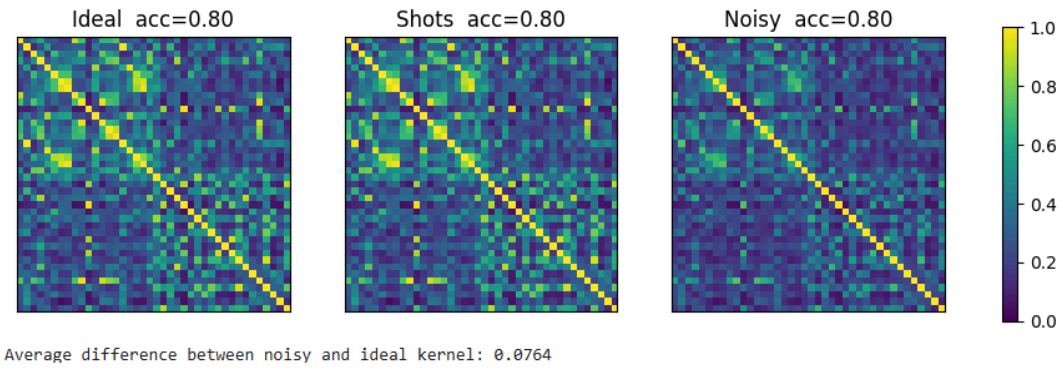


Fig 7. Quantum kernel with no noise, shot noise and simulated IBM device noise

Running on Real IBM Quantum Hardware

We ran the same kernel on ibm_kingston, a real 156-qubit IBM processor, using 500 shots per circuit with error suppression turned on. The hardware run also gave 80% test accuracy, the same as the ideal and noisy simulations. The kernel measured on hardware had a 0.99 correlation with the ideal kernel. Noise lowered all similarity values by about the same amount (from 0.34 to 0.26 on average), but kept their order, which is what the classifier relies on. Since this test used only 20 test objects, it shows that the hardware matches the simulation, not that it is better.

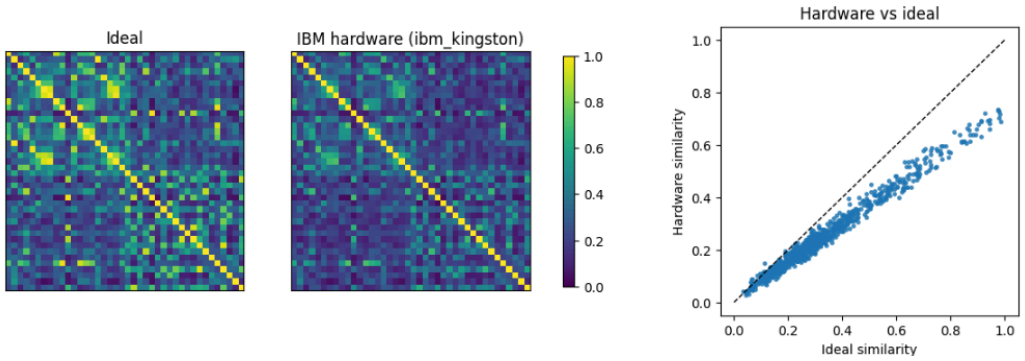


Fig 8. Ideal kernel (left), kernel measured on IBM ibm_kingston (middle), and hardware vs ideal values (right)

Breakup Attribution

We used the quantum kernel to identify which of four breakup events a debris fragment came from:

- Events: Fengyun-1C (2007 anti-satellite test) [7], Cosmos 2251 (2009 collision with Iridium 33) [8], and two Long March 6A rocket stage breakups (2022 and 2024) [9,10]
- QSVC accuracy: 88.2% (random guessing: 25%)
- Classical models: SVM 88.5%, random forest 89.6%, logistic regression 80.9%

- Result: the quantum kernel performed close to the best classical models and gave the most consistent results across samples

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PARENT
Fengyun-1C (2007 ASAT test)     1904
CZ-6A stage (2022 breakup)       612
CZ-6A stage (2024 breakup)       571
Cosmos 2251 (2009 collision)     569
Name: count, dtype: int64

Breakup attribution test accuracy (random guessing = 0.25):
                      mean    std
model
Logistic regression  0.809  0.062
QSVC (tuned)         0.882  0.025
SVM (tuned)          0.885  0.029
Random forest        0.896  0.054
```

Fig 9. Breakup attribution test accuracy for quantum and classical models (random guessing = 25%)

Conclusion

In this hackathon we applied quantum machine learning to a real space debris problem and learned that a fair test matters as much as the model itself. On our first sample, the QSVC (86.67% test accuracy) beat a default classical SVM, but once we filtered the data and tuned every model, quantum and classical models performed about the same (around 89% to 91%). We found that the data's scaling *(the bandwidth)* decides whether a quantum kernel works at all, and that the kernel holds up well under noise and on real IBM hardware. We also saw that unsupervised methods can reveal hidden structure in the data, such as the high-orbit group, and that the same quantum approach extends to new problems like identifying which breakup a debris fragment came from. We did not find a quantum advantage on this task, but we learned how to build, test and compare quantum models honestly, from simulation to real IBM hardware.

Sources

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